

QINGLONG, China

WMO: 544360

Lat: 40.4087N

Lon: 118.9522E

Elev: 838

StdP: 14.26

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 1.4	(c) 4.6	(d) -19.3	(e) 2.0	(f) 15.9	(g) -15.3	(h) 2.5	(i) 16.8	(j) 15.2	(k) 24.8	(l) 12.9	(m) 25.4	(n) 1.4	(o) 220	(p) 0.420

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	91.1	71.2	88.7	70.9	86.5	70.6	78.4	85.0	76.8	83.0	75.3	81.2	6.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.5	142.5	82.3	75.1	135.7	80.6	73.5	128.8	79.5	42.8	85.1	41.1	83.4	39.6	81.3	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.9	12.6	11.0	DB	-4.2	95.9	4.6	3.0	-7.4	98.0	-10.1	99.8	-12.7	101.5	-16.0	103.7
			WB	-5.0	80.3	4.3	1.5	-8.1	81.4	-10.6	82.3	-13.0	83.2	-16.1	84.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	19.9	26.9	39.6	53.8	64.8	72.4	77.1	75.0	65.6	51.6	36.0	23.5
	DBStd	20.92	6.39	7.10	7.55	7.30	5.87	4.56	3.77	4.22	5.66	6.90	7.61	6.44
	HDD50	3263	934	648	334	43	1	0	0	0	63	420	820	
	HDD65	6310	1399	1068	789	341	76	4	0	1	61	417	869	1285
	CDD50	3491	0	0	10	156	458	671	839	776	467	113	1	0
	CDD65	1064	0	0	0	4	69	225	374	312	78	2	0	0
	CDH74	9722	0	0	1	81	922	2155	3329	2541	672	21	0	0
	CDH80	3191	0	0	0	7	249	796	1196	807	134	2	0	0
Wind	WSAvg	3.9	3.1	3.9	5.0	5.6	5.1	4.2	3.6	3.4	3.3	3.6	3.6	3.0
Precipitation	PrecAvg	28.1	0.1	0.2	0.4	1.0	1.7	3.8	9.1	7.7	2.5	1.3	0.4	0.2
	PrecMax	44.4	0.6	1.3	1.1	4.3	5.7	9.2	23.8	20.9	9.1	5.0	1.1	0.9
	PrecMin	18.7	0.0	0.0	0.0	0.0	0.4	0.6	3.4	1.4	0.0	0.0	0.0	0.0
	PrecStd	7.7	0.2	0.2	0.3	1.0	1.2	2.3	4.6	4.7	1.7	1.3	0.3	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.6	52.4	69.1	80.5	90.1	94.7	95.5	92.1	87.0	76.9	61.6	45.3
		MCWB	31.8	38.1	50.2	56.7	63.8	66.4	73.1	73.3	68.1	60.9	48.2	34.8
	2%	DB	38.0	47.7	63.2	76.1	86.0	90.6	91.5	88.9	83.7	71.9	57.4	41.2
		MCWB	28.7	35.7	46.5	55.3	61.6	67.6	73.7	73.3	66.0	57.4	45.6	33.0
	5%	DB	34.9	44.1	58.2	72.5	82.6	87.7	88.8	86.7	80.8	68.4	53.2	38.0
		MCWB	26.9	33.7	43.3	53.3	60.5	67.0	73.6	73.1	65.2	54.9	42.6	30.7
	10%	DB	31.8	40.6	54.0	68.7	79.4	84.7	86.3	84.5	77.9	64.9	49.2	35.0
		MCWB	25.1	31.6	40.6	51.3	59.1	66.5	73.4	72.2	64.1	52.9	40.6	28.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.3	40.4	52.3	61.9	68.8	74.6	80.6	80.1	73.9	63.7	51.0	38.2
		MCDB	40.7	47.7	65.6	74.0	82.2	83.9	87.1	86.3	81.2	71.5	57.5	42.9
	2%	WB	29.7	37.0	47.8	58.6	65.9	72.5	79.0	78.1	70.9	60.9	47.1	34.2
		MCDB	36.4	45.4	60.1	70.7	79.7	82.0	85.8	84.4	78.3	68.8	54.3	39.2
	5%	WB	27.5	34.4	44.5	56.1	63.9	70.8	77.5	76.5	68.6	58.1	44.4	31.4
		MCDB	34.2	43.4	56.9	68.6	76.8	80.3	84.0	82.3	75.5	64.7	51.4	37.1
	10%	WB	25.3	31.7	41.4	53.3	61.8	69.4	76.2	75.1	66.5	55.5	41.4	28.9
		MCDB	31.5	40.2	52.5	65.1	74.0	78.9	82.4	80.5	74.2	62.0	48.7	34.9
Mean Daily Temperature Range	5% DB	MDBR	21.8	21.6	21.3	22.1	23.5	20.4	16.1	17.3	21.7	22.8	20.2	20.5
		MCDBR	24.8	26.3	27.4	28.1	29.9	27.3	21.5	20.3	25.4	27.2	25.4	23.4
		MCWBR	17.2	16.5	15.2	12.5	11.1	9.2	7.5	7.6	10.0	13.4	15.0	16.1
	5% WB	MCDBR	23.6	24.3	25.9	23.5	23.7	19.1	14.4	14.5	18.3	21.5	23.1	21.6
MCWBR		16.5	15.8	15.0	12.2	10.8	8.1	6.7	6.7	8.9	12.4	15.1	15.5	
Clear-Sky Solar Irradiance	taub		0.383	0.466	0.508	0.597	0.640	0.759	0.754	0.674	0.606	0.575	0.448	0.394
	taud		2.040	1.857	1.805	1.675	1.621	1.474	1.520	1.646	1.718	1.721	1.939	2.042
	Ebn at Noon		231	225	232	222	216	192	190	201	204	189	204	215
	Edh at Noon		38	53	62	75	81	94	89	76	67	60	42	35
All-Sky Solar Radiation	RadAvg		820	1093	1491	1718	1930	1722	1569	1564	1411	1092	816	718
	RadStd		56	81	100	95	125	150	136	102	88	81	88	47

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+122	N/A

Nomenclature: See separate page